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Simplifying Creativity with Choices: A User-Centered Comparison of Text-Based and Choices-Based Input Methods in Generative Models

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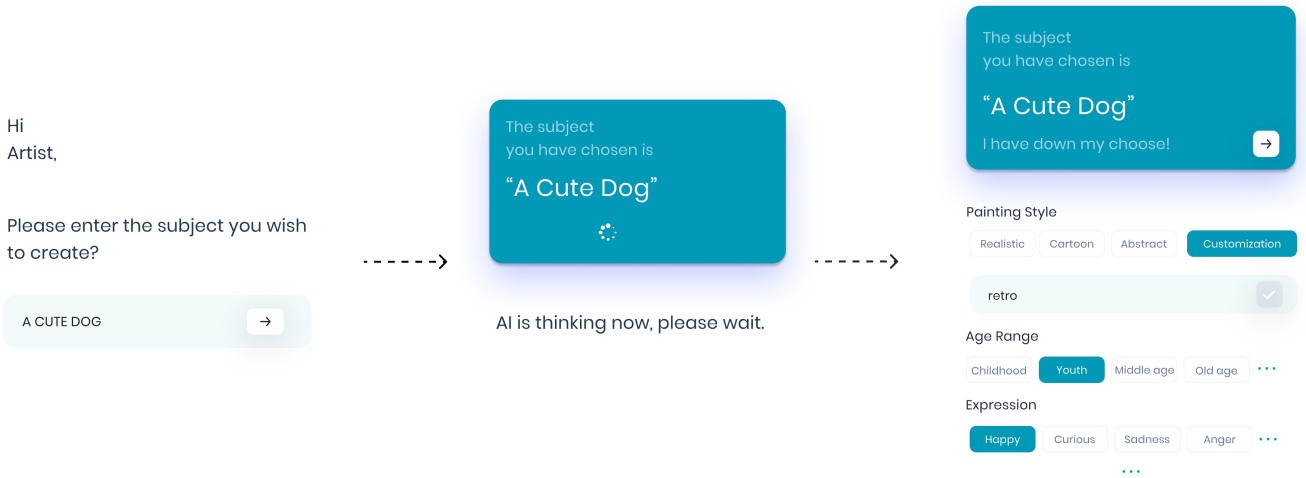

Figure 1: The figure shows the workflow of the Choices-Based input method (Choices2I system) in three steps. First, the user inputs a subject, like "A Cute Dog." Next, the system uses a large language model to process the input and generate detailed attribute categories (e.g., style, age, and expression). Finally, these attributes are presented in a clear UI interface, allowing users to easily select and refine their image prompt.

Abstract

This paper presents a comparison of text-based and choices-based input methods in text-to-image (T2I) systems. While text-based inputs offer flexibility and creative control, they often impose a high cognitive load, especially for novice users unfamiliar with prompt engineering. Therefore, we developed and tested a new system named Choices2I, which offers a structured and choices-based interface. We compared it with the traditional text-based prompts used in GPT-4's official web version. Our results show that users from creative industries (e.g., designers, artists, and film directors) prefer choices-based input (i.e., Choices2I) due to its flexibility and control, regardless of their expertise level. These users often have clear creative goals and find the structured options helpful in achieving specific outcomes. Conversely, non-creative industry users, who typically use T2I models for entertainment or

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casual purposes, favor text-based input because it is simpler and imposes less cognitive load without the pressure of making choices. This research provides critical insights into the comparison of text-based and choices-based input methods for both professional and novice users. These findings yield design insights and guidelines for building more adaptive and user-friendly T2I systems, balancing accessibility for novices with creative flexibility for professionals.

CCS Concepts

• **Human-centered computing → User centered design**.

Keywords

Text-to-Image Generation, User Interface Design, Qualitative Study, Input Methods, User Experience

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1 Introduction

The emergence of Generative Models has introduced a powerful tool in the creative domain, particularly with the advent of Text-to-Image (T2I) models [9, 23, 25, 32, 37]. Since these models can quickly generate complex, beautiful, and detailed images by inputting text, they have gradually become auxiliary tools for content production [9, 32], and applications in various fields (e.g., art, design, education, and entertainment) [7, 26, 31, 34, 35]. Although current generative models have achieved significant progress in model image quality and diversity, designing sensible and effective prompts is a difficult and empirical task. Especially for novice and non-expert users, generating high-quality images requires a specific format, and complex prompt words can be confusing and difficult to start.

Users in most existing T2I systems control images' content, style, and details by entering text prompts. To achieve the prompts required for the target pictures with high quality and quickly, to understand the necessary output clearly, you need to use appropriate prompt keywords, correct prompt structure, and skills to communicate with the LLM [5]. Previous research indicates that for expert users (e.g., prompt engineers, designers, and AI researchers), writing prompts quickly and accurately has significant advantages in generating creative images with a high degree of customization [13]. In contrast, novice users often face substantial challenges in prompt engineering. People unfamiliar with prompt writing in generative models will find it challenging to start and complete effective prompts quickly, it usually requires multiple trial and error cycles to get close to the desired results [3]. In addition, this iterative prompt adjustment process will bring a high cognitive load and increase the learning curve, making it difficult for T2I technology to be widely adopted in various fields [11].

To address the current challenges, many research works and system designs aim to simplify the interaction between users and T2I models, for example, stable diffusion [25] and commercial models such as MidJourney[1] and Dall-E 3[2] demonstrate progress in this field. Some studies introduce features such as prompt suggestions and natural language interaction to assist the input process [3]. However, most of the current solutions are based on optimizing or assisting the text input process, which is ideal for expert users but is still not intuitive and simple enough for novice users who are not familiar with how generative models work or lack experience with AI (e.g., elderly users, digital novices, etc). For them, quickly writing effective prompts is still a problem that needs to be solved.

This study proposed a new interaction method, the choice-based input method, to address these issues. This method provides users with real-time generated attributes (e.g., picture style, scene elements, etc) for selection based on a large language model. It lets them quickly write high-quality prompts and generate images by selecting and clicking without writing complex text prompts. It could reduce the user's cognitive burden by providing easy-to-understand options, making it easier for novices to accept the interaction. In addition, we developed the Choices2I system to implement this interaction method, allowing users to generate images through a structured process (see Figure 1). Although in theory, this system can reduce the learning curve and improve usability for non-experts, systematic user studies are still needed to verify its effectiveness in different user groups.

This study aims to qualitatively compare text-based (ChatGPT) and choice-based (Choices2I) input methods in T2I systems. We ask participants to use both systems and complete a series of tasks, and feedback from different types of users (experts and novices) was obtained. Through participant interviews and questionnaire surveys, we studied the effects of these input methods on their satisfaction, cognitive load, and creative expression. Before the user experiment, it was assumed that expert users would prefer text-based input because of its flexibility and control. In contrast, novice users prefer choice-based methods because of their simplicity and low cognitive load. However, the results after the experiment revealed different results. Several participants from the creative industry (both expert and novice users) preferred the choice-based method over the text-based method, and they believed that the system provided professional-level options that could enhance their creativity. On the other hand, novice users who lack a clear purpose may find the choice-based method cumbersome in the creative process. Although the generated content became more specific, users who only tried it did not like this experience. Overall, based on the preliminary results of this study, our contributions are mainly in the following aspects:

(1) **Design and implementation of the Choices2I system:** A choice-based interaction method was developed and implemented, allowing users to generate images by selecting attributes provided by the system (e.g., style, age, and expression). This method effectively reduced the cognitive load of users and simplified the process of text-prompted input.
(2) **Qualitative analysis comparison of text-based and choice-based input methods:** A qualitative analysis was conducted

[1] www.midjourney.com
[2] openai.com/index/dall-e-3

to understand the experience of different types of users (experts and novices) when interacting with the T2I model using text-based and choice-based input methods. The results showed that participants from creative industries preferred choice-based input because of its flexibility and control. In contrast, participants from non-creative industries preferred text-based input because of its simplicity, low cognitive load, and lack of choice pressure.

(3) **Understanding the potential preferences of creative industry users for choice-based input:** Although choice-based input methods are generally considered more suitable for novices during the design and hypothesis stages, some users from creative industries reported that choice-based input can provide professional-level options that inspire their creativity. This finding challenges previous assumptions and introduces a new perspective, emphasizing the need to design diverse input methods from the perspective of different professional user groups rather than simply categorizing T2I users as novices or experts.

(4) **Design guidelines for future T2I interfaces:** From the user study, we propose five design recommendations to provide more adaptable and user-friendly interfaces for various professional user groups, especially those in creative industries. These guidelines provide new design insights and theoretical foundations for the future development of systems based on choice-based input methods, balancing accessibility and creative flexibility.

2 Related Work

2.1 Text to Image Generation Models

T2I model is a type of generative model that refers to the process of utilizing textual descriptions as input to generate corresponding visual images [6, 16, 17, 21–23, 25, 36]. One of the pioneering models, AlignDRAW, often produced blurry images and lacked photorealism [16]. OpenAI's DALL-E [22] emerged as one of the first models to receive widespread attention. Following the success of diffusion models [1, 21, 25, 27, 33], demonstrated remarkable abilities in image generation. DALL-E 2 [21], with its advanced capabilities, produces more intricate and realistic images, integrating a diffusion model and pre-training with CLIP for augmented functionality [20]. However, a universal challenge persists with these models: it becomes essential for users to acquire skills in crafting prompts with greater efficacy, which involves significant trial and error. To address this, DALL-E 3 [2] introduced a more conversational approach by integrating with ChatGPT, allowing users to refine prompts through natural language interaction. However, conversational interfaces rely heavily on linguistic input, posing barriers for novice users. Alternative approaches like Promptify [4] and 3DALL-E [14] highlight the role of UI designs in reducing cognitive load, though they often lack flexibility or adaptability. Our proposed choices-based interface builds on these insights, dynamically generating pre-selected attributes (e.g., styles, elements) to simplify the process, enhancing accessibility for all users.

2.2 Prompt Engineering

In the realm of natural language processing, a prompt is a structured textual input that steers a language model towards generating targeted outputs [12, 18]. Prompt engineering refers to the method of crafting text prompts in a manner that aligns with the model's understanding, thereby guiding the generation process towards the intended results [24, 28–30]. At present, a significant portion of research in prompt engineering is focused on text generation within natural language processing, especially those studies that utilize text as an input for generating image outputs, commonly referred to as T2I. The advent of T2I models like DALL-E, SD, and Midjourney has catalyzed a significant uptick in the application of generative models across diverse fields, consequently elevating prompt engineering to a position of widespread prominence. In contrast to typical LLMs that are adept at understanding grammar and sentence structure, T2I models require a unique set of prompting techniques. During image generation, the emphasis is on inputs that succinctly describe the subject and style (e.g., "a small dog in the style of hyperrealism"), rendering complex natural sentences and the intentional inclusion of connecting words generally superfluous. Our proposed system guides users in choosing appropriate textual inputs through an innovative T2I interface, eliminating the need for novices to learn these complex prompt techniques deliberately. Furthermore, we employ LLMs to augment the image's details, thereby ensuring a consistent and precise representation of the user's envisioned outcome.

2.3 User Interface for T2I

User Interfaces (UI), as forms of visual feedback and interaction, are increasingly integrated into generative models and their various applications, highlighting the growing attention towards UI in these contexts [8]. Opal [13], a human-AI co-creative system designed for text-to-image generation in news illustration, improves user experience and generation outcomes. 3DALL-E, a plugin integrating DALL-E, GPT-3, and CLIP within CAD software, generates 2D image inspirations for 3D design, empowering users to create text and image prompts tailored to their modeling projects. One work with Promptify offers a more efficient browsing experience than the conventional folder view by introducing an innovative interface [4]. Mini DALL-E 3 [11] introduces the task of interactive text-to-image using a novel, natural language-based human-machine interface, enhancing T2I system capabilities. Building on these existing systems, our proposed T2I interface leverages LLM to interpret and contextualize user intentions for enhanced image generation dynamically. Transforming selections into optimized textual prompts might reduce users' workload and enhance the overall user experience.

3 Choices2I System Workflows

The Choices2I workflow consists of three key stages: *Prompt embedding*, *Text parsing*, and *Selection aggregation* (Figure 2). It starts when the user inputs a central theme describing the desired image. The system first processes this input through the *Prompt Embedding* phase, where the text is contextualized within a set of predefined prompts. This embedded prompt is then passed to an LLM, which generates a range of customized attributes and options that align

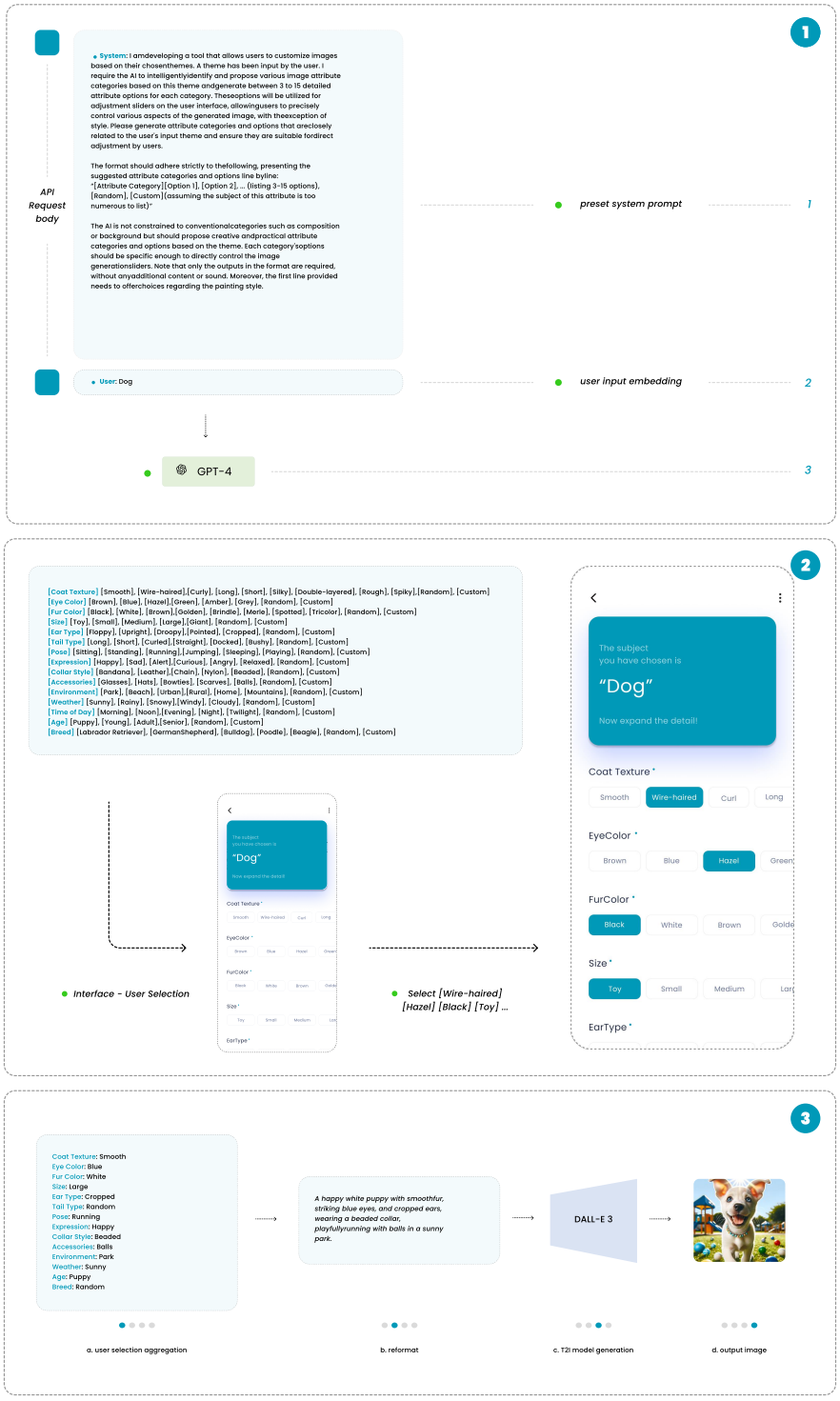

Figure 2: 1) The main request body when using OpenAI's ChatGPT API; 2) When the input is "Dog", the LLM processes and the resulting text output is converted into a user interface that the user can select; 3) The process of user selection aggregation and reformat.

with the user's theme. Next, the output from the LLM undergoes a critical *Text Parsing* phase, where the system interprets the generated content and translates it into user interface elements. This step allows users to explore and select from various options intuitively. Once the user has made their selections, these choices are combined into a cohesive text prompt called *Selection aggregation*. Depending on the input requirements of the image generation model, this prompt is either further processed by the LLM or directly fed into a T2I model.

This framework includes user input, context embedding, and AI processing to simplify the input steps of generating visual content. It increases the system's ease of operation and reduces the difficulty of writing prompts so that users can create images with minimal technical knowledge. The following sections will explain each component in detail.

3.1 Prompt Embedding

After the user inputs the central theme, the system will expand the topic details through *Prompt Embedding*. From an implementation perspective, in order to balance processing delay and performance, we use GPT 4-Turbo as the LLM model in our system. In the API, two key parameters make up the language settings of LLM: "system" and "user". The "system" defines the system's response settings, while the "user" is the content of each conversation. In our setting, the context of the "system" is shown in Figure 2, and the input of the "user" is composed of the keywords selected by the user for the description of the scene.

The system prompts that drive LLM to generate options revolve around several cores: 1) The topic entered by the user is used as the basis for generation; 2) The image attribute categories related to the topic must be intelligently identified; 3) LLM will generate 3 to 15 specific options under each attribute category; These options must be detailed enough; 4) The output format is required to facilitate processing in the *Text Parsing* step.

3.2 Output Text Parsing

Once the Prompt Embedding phase is complete, the raw output from LLM is processed. As shown in Figure 2 (using "Dog" as an example), our back-end architecture organizes the LLM output into clearly defined titles and interactive elements (e.g., buttons for the front-end interface). This structured output allows users to make selections from various categories.

The interface provides "Not Defined" and "Custom" options to accommodate different user preferences. The "Not Defined" option lets users leave certain attributes open-ended. In contrast, the "Custom" option offers a text field for personalized input to enhance user control over the output, allowing for structured and personalized image generation.

3.3 User Selection Aggregation

After the user makes their selections, the system aggregates these choices into a structured input prompt for the T2I model, which enables the T2I model to effectively interpret the selections and generate an image that reflects the user's intent.

Choices2I is designed to be flexible and adaptable to various T2I models by allowing customization of the aggregation methodology for compatibility with different models, enhancing the accuracy and relevance of the generated prompts. Therefore, the system can meet diverse user requirements and be tailored to the specific T2I prompt structure.

Our system uses DALL-E 3 as the T2I model (Figure 2), which can comprehend natural language prompts without relying on traditional T2I models, like the stable diffusion model's positive and negative prompt structure. Thus, the final prompt is the direct input to the image generation model, producing the visual content.

4 Method

The primary objective of this study is to conduct a comparative analysis of the choices-based input method (Choices2I) against the traditional text-based input method (OpenAI's official ChatGPT web version includes DALL-E 3). Specifically, we aim to assess whether Choices2I can lower the entry barrier for novice users and stimulate creativity for professional users. Ultimately, the goal is to determine if this system can quickly and effectively meet user needs.

In this study, participants were tasked with two platforms: OpenAI's official ChatGPT web version, which includes DALL-E 3(referred to as ChatGPT-Web), and Choices2I. The interface is familiar to most users and has been enhanced for accessibility through its integration with ChatGPT, broadening its appeal to casual or first-time users, so it's perfect as a model for Choices2I comparisons. User study comparative evaluation and semi-structured interviews in our study to comprehensively understand user experiences with these T2I models. The ChatGPT-Web and Choices2I were utilized on computer browsers to ensure an equitable comparison. This setup maintained consistency in user interaction, with all participants navigating both systems using a keyboard and mouse, except for some users who are unfamiliar with the keyboard and will use the voice input method to type. We randomly divided the participants into two equal groups to mitigate any potential bias from the order in which the system was used. One group commenced their experience with Choices2I, while the other began with the ChatGPT-Web interface.

Experimental procedure All participants were given a detailed explanation of the process and operation by the researchers before starting the experiment. They were taught how to use both user interface systems, ensuring they could use the system themselves. Under the guidance of the experimenter, there were three experimental tasks for the user's interactive experience. The purpose of these tasks was to evaluate participants' ability to describe concrete and abstract objects and generate images based on familiar, personally favored subjects.

The participants need to think of a **favorite animal** first, then use the system to generate a picture of it. Secondly, participants are asked to think of a **favorite seasonal landscape** and then generate it. Finally, participants were **free to experience** the system and generate whatever they liked. After examining all three topics, the participants will fill out a questionnaire about the system. A semi-structured interview will then be conducted, and the interview results will be recorded after both systems have been experienced.

Table 1: Participant Information N=16

ID	Gender	Age	Occupation	Prior T2i Experience	Prefenrence
1	Male	35	Designer	Some	Prefers C2I.
2	Female	25	Student	None	Prefers T2I.
3	Male	28	Designer	Occasional	Prefers C2I.
4	Female	30	Researcher	None	Prefers C2I.
5	Male	30	Engineer	None	Prefers T2I.
6	Male	54	Retired	None	Prefers C2I.
7	Male	27	Student	None	Prefers T2I.
8	Male	29	Engineer	None	Both.
9	Male	35	Designer	None	Prefers C2I.
10	Female	32	Film Director	None	Prefers C2I.
11	Male	40	Storyboard Artist	Limited	Prefers C2I.
12	Male	35	Designer	Extensive	Prefers C2I.
13	Female	25	Student	None	Prefers C2I.
14	Male	28	Designer	Some	Prefers C2I.
15	Male	35	Engineer	Limited	Both.
16	Female	29	Artist	None	Prefers C2I.

4.1 Participants

We recruited 16 participants (11 males and 5 females; M = 32.31, SD = 6.91) for the experience, among whom 7 participants had prior experience with AIGC systems, while the remaining 9 participants were novices in this domain. All of them are instructed to generate images using the ChatGPT-Web and Choices2I, and the order of use for each participant is randomized. First, the experimenter will explain the use of the system, and then about 20 minutes will be used to generate images of the three tasks, and the questionnaire will be filled out. After trying both systems, participants were interviewed semi-structured, with an average experience time of about 50 minutes per participant.

4.2 Quantitative Evaluation

Participants have approximately 20 minutes to experiment with each system independently. Once completed their trial with both systems, participants were asked to complete a survey. The participants' basic information, including age and familiarity with generative AI, was recorded at the beginning of the questionnaire. These questionnaires assessed users' workloads, user engagement, ease of use, and system usability. We selected these three questionnaires because they collectively comprehensively evaluate critical and representative aspects of the user experience. ***NASA-TLX Assessment*** We employed the NASA Task Load Index [10] on a 1-7 Likert scale to measure six user experience facets: mental demand, physical demand, temporal demand, performance, effort, and frustration level, explicitly tailored for T2I context. This index is validated by extensive use in other T2I user interface research. ***User Engagement Scale Short-Form*** UES-SF [19] using a 1-7 Likert scale, evaluated focused attention, usability, aesthetic appeal, and the sense of reward. ***USE Questionnaire*** The USE Questionnaire [15] uses a 1-7 Likert scale to assess the system's usefulness, ease of learning/use, and overall satisfaction. We used short forms, then translated the Chinese versions to avoid the numbness of participants due to excessive problems.

4.3 Semi-structured Interview

After participants completed both tasks, they were invited for an eight-minute interview. They had to answer questions asked by the researchers and then tell how they really felt during the experiment. Participants' feedback will further help to improve the experiment process. These interviews probed deeply into various aspects: participants' preferences between the two interfaces, their perceived ease of use, the relevance of our system's provided options, and potential areas for improvement. Five questions were: 1) Which interface do you prefer: Choices2I or the text-based system, and why? 2) In your opinion, does the Choices2I offer more advantages over the text-based method in a text-to-image system context? 3) How well do the expanded options align with your expectations

and creativity? Does an additional image element enhance your experience? 4) Do you think Choices2I effectively lowers the barrier to entry for new users of text-to-image systems? 5) What specific improvements or features would you suggest to enhance our system's functionality and user experience?

5 Results and Findings

5.1 Quantitative Analysis

NASA-TLX In the analysis of the impact of two user interfaces (ChatGPT-Web and Choices2I) on the workload, a paired samples t-test was conducted. The results indicated a significant difference. Specifically, the mean score of workload under ChatGPT-Web ($M = 4.01$, $SD = 0.69$) was significantly higher than under Choices2I ($M = 3.29$, $SD = 0.79$), $t(15) = 2.32$, $p < 0.05$. This suggests that using ChatGPT-Web exerts a more substantial influence on the users' workload than using Choices2I does.

User Engagement A statistical examination was carried out to discern the differential impact of using ChatGPT-Web and Choices2I on the outcome referred to as user engagement. This inquiry employed a t-test for paired samples to establish any significant disparities. The findings revealed that the impact on user engagement was not uniform across these two user interfaces. It was observed that the average score when subjected to ChatGPT-Web ($M = 5.15$, $SD = 0.72$) surpassed that of Choices2I ($M = 5.86$, $SD = 0.57$), $t(15) = -2.36$, $p < 0.05$.

USE Questionnaire In the examination of USE across usefulness, ease of use, ease of learning, and satisfaction, we aimed to discern whether two distinct user interfaces, ChatGPT-Web and Choices2I, differed in their impact on user experience. The results from the paired samples t-tests revealed that, in terms of perceived usefulness, users did not statistically favour one interface over the other, with ChatGPT-Web scoring an average of ($M = 4.77$, $SD = 1.40$) and Choices2I scoring ($M = 5.56$, $SD = 0.90$), $t(15) = -1.75$, $p = 0.11$. Similarly, no significant difference was found in ease of use between ChatGPT-Web ($M = 5.12$, $SD = 1.11$) and Choices2I ($M = 5.42$, $SD = 0.72$), $t(15) = -0.76$, $p = 0.46$, or in ease of learning with ChatGPT-Web ($M = 6.08$, $SD = 0.56$) compared to Choices2I ($M = 6.52$, $SD = 0.83$), $t(15) = -1.93$, $p = 0.08$. However, the dimension of satisfaction revealed a notable divergence; users reported a significantly higher satisfaction with Choices2I ($M = 5.85$, $SD = 0.71$) relative to ChatGPT-Web ($M = 4.96$, $SD = 0.83$), $t(15) = -2.70$, $p = 0.02$. This suggests that while the interfaces may be comparable in terms of their utility, ease of use, and learnability, there is a distinct preference in satisfaction towards Choices2I. This finding underlines the importance of satisfaction as a standalone factor in user experience, potentially influenced by elements beyond those captured by the other three dimensions. Future research could benefit from exploring the specific aspects of user interfaces that contribute most significantly to user satisfaction.

5.2 Qualitative Analysis

Our initial hypothesis was that expert users would prefer text-based input due to its flexibility and control, while novice users would favor the choice-based method for its simplicity and reduced cognitive load. However, the results revealed a more nuanced reality. Through qualitative interviews, we gathered a range of insights into the user experience of both input methods, uncovering unexpected preferences and challenges across different user groups.

5.2.1 Expert Users' Preference for Choices-Based Input. Contrary to our hypothesis, several expert users (mainly from creative industries) strongly preferred the choice-based input method. P1, an experienced AIGC user and designer, highlighted that the Choices2I system offered professional-level options that significantly enhanced his creativity. He explained that the structured options allowed him to explore ideas he had not originally considered, thus pushing the boundaries of his creative process. The system's ability to provide curated, expert-level suggestions was seen as a major advantage, as it saved time in ideation and helped experts from creative industries like P1 quickly form new creative concepts. P11, a storyboard artist, echoed this sentiment, noting that while text-based input allowed for greater flexibility, the abstract nature of text descriptions often made it challenging to get started. In contrast, the choices-based system provided a clear set of options that acted as a "guide," helping him ease into the creative process without needing to have a fully formed idea at the outset. P11 is not considered an expert in the AIGC field, as he has only limited experience with T2I models. However, like P1, P11 is also a professional in the creative industry, both having rich backgrounds in creative work. This enables them to evaluate the potential and limitations of these tools from a professional perspective, thereby contributing a fresh viewpoint to the research. This hints at a broader theme: **creative industry practitioners, whether experts or novices, tend to prefer structured guidance in their creative process.**. The Choices2I system, by offering professional-level options, not only simplifies the often complex ideation process but also provides a clear sense of direction. This finding challenges our previous assumption that expert users would uniformly prefer the freedom of text-based inputs. Additionally, when faced with complex and professional-level tasks, many experts from the creative industry found that Choices2I helped them achieve high-quality results more efficiently. Experts noted that although they possess high levels of expertise, the structured options allowed them to more effectively push the boundaries of their creativity, saving time and improving the overall quality of their work. Choices2I transformed what could have been a laborious ideation process into one of guided exploration, making it easier for them to achieve high-quality results.

5.2.2 Novice Users' Frustrations with Choices-Based Input. In contrast, novice users did not show the expected preference for the choice-based method. While we anticipated that choice-based input would reduce cognitive load for less experienced users, some novice participants found the process burdensome and limiting. For example, P6, a participant unfamiliar with advanced technology, initially appreciated the reduction in typing required by the choices-based system but expressed frustration later because the predefined options constrained creative freedom. The participant wanted to explore more freely and described the system's structure as "restrictive," Similarly, P7 found that although the choices provided by the system helped him clarify his ideas, they also made the creative process feel more tedious and expressed dissatisfaction with the rigid structure of the choices, which forced him to make decisions at every step, leaving little room for spontaneous creativity. The sentiment among novice users like P6 and P7 was that

	ChatGPT-Web (M, SD)	Choices2I (M, SD)	t(15)	p-value
Workload (NASA-TLX)	4.01 (0.69)	3.29 (0.79)	2.32	< 0.05
User Engagement	5.15 (0.72)	5.86 (0.57)	-2.36	< 0.05

Table 2: Comparison of workload (NASA-TLX) and user engagement between ChatGPT-Web and Choices2I

Dimension	ChatGPT-Web (M, SD)	Choices2I (M, SD)	t(15)	p-value
Perceived Usefulness	4.77 (1.40)	5.56 (0.90)	-1.75	0.11
Ease of Use	5.12 (1.11)	5.42 (0.72)	-0.76	0.46
Ease of Learning	6.08 (0.56)	6.52 (0.83)	-1.93	0.08
Satisfaction	4.96 (0.83)	5.85 (0.71)	-2.70	0.02

Table 3: Comparison of user experience between ChatGPT-Web and Choices2I across four dimensions: Perceived Usefulness, Ease of Use, Ease of Learning, and Satisfaction.

while the system made the content more specific, it did not necessarily make the creative process more enjoyable. This feedback highlights the importance of balancing structure with freedom in innovative systems. Although choice-based input was designed to simplify the process for novice users, it would introduce a new form of cognitive load—decision fatigue—that some users found overwhelming. And it is worth noting that P6 and P7 are not practitioners in the creative industries.

5.2.3 Decision Fatigue and Option Overload. Both novice and expert users had the issue of decision fatigue. P4 and P5 found that while the choices-based system helped them quickly identify the content they liked, the number of options presented could be overwhelming. P4 says that while he generally did not mind making choices, the count of options sometimes became tiresome, especially when faced with several decision points in succession. This issue was particularly acute for novice users, who expressed that the abundance of choices sometimes made the system more complicated than helpful. P4 suggested that the system could present smaller options first, allowing users to expand choices as needed. So, the system should provide a more human-centric design that balances simplicity and flexibility.

5.2.4 The Need for Dynamic and Customizable Options. Several participants, P2, P3, and P12, emphasized the importance of dynamically generated options that adapt to user input. They say the current system lacks the flexibility to respond to more detailed creative needs. P6, for instance, mentioned that he wanted to adjust specific elements—such as the characters' posture in the images—but the system did not provide sufficient options for such fine-grained control. In summary, novice and expert users suggested that the system should allow for more customization and dynamic generation of options based on user input rather than relying solely on predefined choices. So, a key limitation of the current choices-based system: **it excels in providing structured guidance for broad creative tasks but struggles to accommodate more detailed, specific user needs**.

5.2.5 Complex Themes and Option Relevance. Another challenge observed among expert users was the handling of complex themes. P1 found that when working on a complex project involving a "subtropical rainy scene," the system generated some irrelevant options that did not align with his creative needs. He suggested that Choices2I might perform better for more straightforward tasks or vocabulary expansion, but it struggled to provide relevant options when dealing with more intricate themes. P7 noted that the system sometimes offered unexpected, creative options, but not all were suitable. So, Choices2I needs to refine its option generation process and algorithm, especially when dealing with complex concepts, so that users can make more contextually relevant choices.

5.2.6 Professionalism and User Understanding. Expert participants appreciated the professionalism of the provided options. P8 says that specific options, although unfamiliar, added a level of sophistication to the generated content. However, both P8 and P12 also pointed out that these more specialized options could confuse novice users who lack the background knowledge to understand them. For the solution, P12 suggested adding explanations or visual tips to help users, especially those less familiar with artistic terminology, understand the meanings of specific options. Participant feedback suggested that **while expert users find specialized, complex options more valuable, novice users need additional help to benefit fully**. Participants felt that explaining keywords could fill this gap and enable novice users to use the system's advanced features more effectively.

5.2.7 Multi-Selection and Post-Generation Refinement. Several users, including P3 and P11, expressed a need for multi-selection functionality in specific dimensions, such as character accessories. They found that being limited to a single choice per dimension restricted their ability to customize the output fully. Moreover, P6 and P11 suggested that the system should allow users to revise their choices after viewing the generated image, enabling more iterative refinement. This aligns with P11's desire to continue adding new options even after the initial image was generated, making the system more flexible for ongoing creative exploration.

6 Design Guidelines for Future T2I Interfaces

Based on the qualitative insights, we propose design guidelines to develop more adaptive and user-centric T2I interfaces. These guidelines emphasize using choice-based input methods to address the

distinct needs of novice and expert users, with particular attention paid to professionals in creative industries. By prioritizing system flexibility, the guidelines ensure that the interface accommodates varying levels of expertise, thereby supporting diverse creative workflows and enhancing user experience.

6.1 Offer Text-Based and Choices-Based Input Modes Both

One of the striking findings was the diverse range of user preferences for input methods. Contrary to our initial hypothesis, some expert users (e.g., P1 and P11) strongly preferred the choice-based method due to its creative guidance. In contrast, several novice users reported the choice-based system to be restrictive and cumbersome. This points out that no single input method is universally superior—users need options to choose the method that fits their approach to creativity.

Guideline 1: The system should provide both text-based and choice-based interaction methods, making it easy for users to choose the method that best suits their task or preferences. For example, text-based input can provide more freedom and fine-grained control. In contrast, choice-based input can provide structured support, especially for novice or experienced users who need to get results quickly.

6.2 Interaction methods with increasing difficulty

Although novice users believe that the choice-based method can reduce the difficulty of use, some feedback suggests there is decision fatigue and limited creativity. For example, P7 and P6 mentioned that although choice-based input helps them to sort out their thoughts, it also makes the process boring. Novice users are more accustomed to a step-by-step learning route or operation path. There will be a negative effect if too many choices are given in the initial stage.

Guideline 2: Design a step-by-step interaction method for novice users. The initial stage of the system is a simpler interface with fewer choices. Still, as the user's familiarity increases, the complexity and diversity of the options can gradually increase, even based on text prompt input. In addition, to reduce users' difficulty understanding the options, prompts or visual aids for complex choices should be added, as suggested by P12.

6.3 Provide more professional input choices for expert users

Although the original design goal of this system was to lower the threshold for novices, when used by experts, the choice-based input received feedback that professional-level options can increase creativity. For example, P1 and P11 emphasized that the professional options generated by the system inspired them to go in directions they had not considered during their thinking process. This shows that even expert users can benefit from structured guidance, encouraging exploration and stimulating creativity in complex tasks.

Guideline 3: Continue to provide expert users with professional choice-based input options that are highly flexible and precise. For example, expert users should be able to select multiple options or collaboratively adjust their choices with AI. This will make the system more suitable for complex tasks, such as multi-layer design projects, especially for fine-tuning more fine-grained settings.

6.4 Reduce decision fatigue by balancing the number of options

Both novice and expert users said they experience decision fatigue when presented with too many options. For example, P4 and P5 noted that while various options are useful for inspiration, too many can be frustrating and reduce engagement. The trade-off is between providing enough options to stimulate creativity and avoiding too many options.

Guideline 4: Balance the count of options presented to users at each creative process stage. For novice users, start with a smaller, broader set of options and expand the selection as needed. Expert users should consider providing multiple selection capabilities for more complex dimensions, such as character accessories or style elements, as suggested by P3 and P11.

6.5 Dynamic generation of contextual options

Some users suggested more flexible options for expansion. For example, P2 and P12 thought that the system only generated options once, which made the user's choice too rigid, and many options did not match expectations. Thus, a more flexible system that can be expanded in real-time or regenerated is needed.

Guideline 5: Implement dynamic and contextual option generation in choice-based input systems. Avoid limiting users to a fixed set of options. Novice and expert users can access more relevant and task-specific options as their creative goals evolve. This helps provide users with options that match their creative direction and reduce frustration.

7 Discussion

7.1 Structured Professionalism in Creative Workflows: Balancing Guidance and Flexibility

Feedback from the participants of this study indicated that our proposed Choices2I helped to generate images in a structured and professional manner compared to traditional text input methods. According to the study, they were surprised that this system was able to guide them through a systematic process similar to a professional project management workflow for creative content. Moreover, we found this approach to be particularly effective in helping users manage complex creative tasks, especially by minimizing the ambiguity that is often associated with open-ended text prompts. At the same time, these qualitative results highlight the importance of striking a balance between user freedom and structured guidance. Some of these participants noted that the system's predefined choices allowed them to focus on refining and improving their creative output itself, rather than being entangled in the complexities of prompt engineering and thus losing patience. This suggests that a more structured input approach, as exemplified by Choices2I, can help users, especially those with less expertise, to achieve high-quality results with minimal cognitive burden. However, this also

raises some questions about the need to balance structure and flexibility. Some users expressed a desire for more adjustable options, suggesting that pre-defined choices, while useful, can occasionally feel limiting.

7.2 Cognitive Load and Workload Reduction

NASA-TLX data confirm that our proposed Choices2I significantly reduces cognitive workload compared to traditional text-based input methods. Participants in the interviews often indicated that generating descriptive text prompts requires a great deal of mental effort, especially when trying to express complex ideas. Text-based approaches often force users to think deeply about how to translate their ideas into coherent and concrete prompts, which undoubtedly increases their cognitive load. In contrast, Choices2I simplifies this process by providing predefined options that allow users to bypass much of the cognitive load associated with crafting precise text prompts. In addition, we found that this reduction in cognitive load was particularly evident among new users, who found the simplified selection process more intuitive and less mentally taxing. As one participant noted, "The choices make it easier for me to click and see what happens instead of struggling to think about what I want to say." This feedback matches objective data from NASA-TLX, which shows a significant reduction in workload when using Choices2I. Thus, we conclude that Choices2I, by minimizing the cognitive effort required to generate high-quality images, provides an easier entry point for users who may have difficulty coping with complex textual input, including those less familiar with the technical aspects of the T2I model.

Interestingly, the reduced workload doesn't just benefit novice users. Even expert users, who are usually more adept at working with textual prompts, appreciated the reduced mental burden of choice-based input proposed by the Choices2I system. In addition, they were pleasantly surprised to find that Choices2I allowed them to generate creative ideas quickly, making it particularly suitable for early-stage ideation and brainstorming. However, the experts also noted that the choice-based approach sometimes lacked the flexibility they desired for more specific or detailed creative tasks (e.g., the system could not be fine-tuned after the options were generated), suggesting that while Choices2I is effective in reducing cognitive load, it may not always provide the depth of control that some users seek.

7.3 User Engagement and Interaction Dynamics

Qualitative feedback on user engagement showed that Choices2I was significantly more interactive and engaging than traditional text-based systems such as ChatGPT-Web. Through interviews, participants who took part in the user experiment consistently reported that the choice-based interface made them feel more engaged. They felt that, unlike text-based approaches where users enter text and wait for a response, Choices2I provided an immediate, hands-on interaction that felt more energizing and engaging. This sense of interactivity is very important and is particularly valued by professional users who find text-based approaches too passive and who do not like to wait. Therefore, this interactive nature of Choices2I seems to enhance users' sense of agency, making them feel that they are actively shaping the outcome and that this positivity affects the outcome.

Moreover, we found that the Choices2I system seemed to play an important supporting role in increasing users' creativity. Many participants mentioned that the predefined options provided them with new ideas and inspirations that they might not have considered otherwise. This iterative process of selecting and refining options encouraged users to explore different creative directions, often with unexpected positive results. As one participant put it, "The choices forced me to think outside the box and try things I wouldn't have thought of on my own." However, it is worth noting that the novelty of the Choices2I interface may have contributed to the increased engagement in this study. Several participants noted that the novelty of the system made it more engaging, but they were unsure if this engagement would continue over time (e.g., they became familiar with the interaction pattern and Choices2I interface). Thus, this raises an interesting question for our study about the long-term engagement of users with choice-based interfaces and whether the initial excitement fades as users become more familiar with the system's options.

7.4 Satisfaction and Usability Perceptions

Although one of our design aims was to simplify the authoring process, the USE analysis revealed some user perceptions of its usability that were higher than we expected. In addition, while the system greatly improved user satisfaction, it did not surpass text-based approaches in terms of ease of learning, ease of use, or usefulness. This discrepancy suggests that Choices2I may present a steeper learning curve than initially anticipated, especially for users less familiar with choice-based input methods. Participants noted that while the system provided a more intuitive mode of interaction, some of the generated ranges of options were overly specialized, making them challenging to understand at times. For example, users reported that they were sometimes overwhelmed by the level of detail required by some of the choices, especially when they had not thought about generating some of the other specifics (e.g., after a user entered a puppy, the choices would provide options for lens language, puppy breed, etc.). In this case, the system's structured approach may feel restrictive, leading some users to prefer free and simple text input.

However, despite these challenges, Choices2I still performs well in terms of overall satisfaction. This high level of satisfaction can be attributed to the system's ability to help users generate images that are closely related to their creative expectations. Choices2I makes it easier for users to achieve their desired outcomes by providing more specific options, even though the process itself is not always as straightforward as they would like. It also shows that while Choices2I may require some initial effort on the part of the user to learn, its rewards in terms of satisfaction and creative fulfillment are huge.

7.5 Experiences and Creativity Enhancement of Creative Industry Practitioners

A key finding from this study is that the Choices2I system performed surprisingly better than traditional text-based input methods among creative industry practitioners, particularly in fostering

creativity and improving workflow efficiency. This result challenges our initial hypothesis that a choice-based input method would primarily benefit novice users by reducing cognitive load. Instead, users from the creative sector, such as designers, artists, and film directors—especially those with clear creative objectives—reported that the professional-level options provided by the Choices2I system helped them better articulate their creative intent and stimulated their creativity.

One particularly noteworthy aspect is that the structured choices offered by the Choices2I system enabled these users to quickly define detailed parameters (e.g., style, expression, and scene elements). This process not only reduced the number of iterations required to design the generation prompts but also sparked their interest in exploring different possibilities through the diverse options available. This design advantage is especially significant for creative professionals, who often seek highly customized creative outputs while expecting to generate multiple options within a short period. Moreover, experienced professionals in the creative industry pointed out that the Choices2I system allowed them to focus more on the creative process itself rather than being overly concerned with how to craft the right textual prompts. This "de-textualized" interaction mode is seen as liberating their thought processes, granting them greater freedom in choosing visual elements and thereby facilitating a smoother creative workflow. Therefore, we recommend further development of similar choice-based input systems in future designs, particularly targeting professional users in creative fields, providing them with increased flexibility and control.

8 Limitation and Future Work

The number of options presented can significantly impact the user experience with the Choices2I system. Users may be provided with 3 to 15 different thematic options, influencing their selection process. One issue mentioned is the excessive number of selections required from a user when defining a scene, with feedback indicating that "The number of times I had to make selections was too high, and such fine detail isn't necessary." This specific concern was linked to the detailed input theme of "golden terraced fields in the mountains with farmers working," which prompted the system to refine details for individual elements like mountains, terraces, and farmers. Additionally, it was observed that the system generated too many complex options, causing difficulty for some users. Besides, many users believe that some jargon words can increase the cognitive load during usage. However, some of these terms significantly impact the composition and color of the generated images. Therefore, the choice to use technical terminology should be left to the users. The initial options provided by the system should be more accessible and only control the core content. If users have specific professional or detailed modification requests, dynamically generating content based on these needs would be a better approach. Secondly, this study primarily relies on qualitative interviews to collect data. While this method provides valuable insights into users' subjective experiences, incorporating quantitative experiments would allow for a more precise evaluation of objective measures such as cognitive load, creative output quality, and user satisfaction. Future studies could integrate quantitative analyses and psychometric tools to further validate the effectiveness of text-based and option-based input methods across different user types. Moreover, although the Choices2I system's design offers a structured option-based interface that significantly reduces cognitive load, the generated options are still dependent on outputs from large language models (such as GPT-4). In certain cases, users may feel that the number or variety of options is limited, particularly for professional users with highly personalized needs. Future work could explore how to enhance the diversity and flexibility of option generation, for instance, by introducing more user-customizable options or hybrid input models. This would better accommodate the ease of use for novice users while maintaining creative flexibility for expert users

In our future work, we plan to further design and develop different T2I interfaces tailored to the needs of distinct user groups, including creative industry practitioners, ensuring that both novice and expert users can engage effectively with generative AI systems. Moreover, special attention will be given to designing interfaces that cater to the unique creative workflows of professionals such as designers, artists, and filmmakers, offering them the flexibility and control necessary to enhance their creative processes. For novice users who often find text-based prompt engineering challenging and overwhelming, we aim to develop choice-based interfaces that simplify the interaction process and reduce cognitive load. By presenting predefined options for key image attributes (e.g., style, composition, and expression), novice users can focus on their creative intentions without mastering the nuances of prompt formulation. This structured approach not only aids in lowering the barrier to entry but also provides an intuitive workflow that aligns with their level of experience, allowing them to achieve satisfactory results with minimal frustration. On the other hand, expert users, especially creative industry practitioners, tend to value the flexibility and granular control that text-based prompts offer. The ability to fine-tune and precisely dictate their creative vision through detailed textual inputs is crucial for these users. Text-based systems allow for a deeper exploration of creative possibilities, as experienced users can experiment with complex combinations of descriptors, generating highly personalized and specific outcomes. Therefore, we propose developing hybrid T2I interfaces that allow users to switch between text-based and choice-based inputs depending on their level of expertise and the task at hand.

Finally, although the experimental design of this study considered user performance across different tasks, it did not delve deeply into how task complexity affects input methods. Future research could design more challenging task scenarios to assess the performance differences between text-based and option-based input methods in complex creative tasks, especially in contexts that require interdisciplinary knowledge or higher degrees of creative freedom. These studies would provide clearer guidelines for designing future image generation systems, helping to develop user interfaces that better address the demands of complex tasks.

9 Conclusion

In this study, we introduce a novel Choices-based input method in T2I systems as an alternative to traditional text-based prompts, aiming to address the cognitive load and usability challenges faced

by novice users. After the development and testing of this Choices2I system, we compared two input methods (choices-based and text-based) to assess their impact on user experience for users with varying levels of expertise. Our findings indicate that participants from creative industries (e.g., designers, artists, and film directors) generally prefer the choice-based input method, regardless of their level of expertise. The structured interface of the choices-based approach could offer greater flexibility and control, enabling participants to more effectively achieve their goals of creating clear and high-quality images. These rich predefined options can help them make quick decisions during the prompt generation process, ensuring that specific visual outcomes are achieved. In contrast, participants from non-creative industries, especially novices, tend to favor the text-based input method due to its simplicity. This method reduces their cognitive effort and eliminates the need to choose from multiple options, making it more suitable for entertainment or casual use. Additionally, there are several contributions to our work. We developed and implemented the novel Choices2I system to allow users to generate images through predefined attributes, significantly reducing users' cognitive load associated with complex text prompt engineering. Then, we conducted a qualitative comparison of text-based and choice-based input methods, revealing distinct user preferences across different industries and expertise levels. Our analysis shows that users from creative industries generally prefer choice-based input due to its flexibility and control, challenging the assumption that such methods are only suitable for novices. Moreover, we propose five design guidelines aimed at creating more adaptive and user-friendly T2I interfaces, which cater to the diverse needs of professional users in creative fields by balancing ease of use with creative freedom.

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Appendix: Questionnaire

Basic Information Collection

- Gender: ____
- Age: ____
- Occupation: ____
- Ever used AI-generated images? ____

NASA Task Load Index (1-7 likert scale)

- Mental Demand: In the process of generating the image you want, how much mental and perceptual effort was needed? Was the task easy or demanding, simple or complex?
- Physical Demand: During the image generation process, how much physical effort was required? Was the task easy or demanding, light or intense, peaceful or strenuous?
- Temporal Demand: While generating your desired image, how much time pressure did you feel? Was the pace slow and leisurely or fast and frantic?
- Performance: How do you rate the success in accomplishing the goals of the task, and how satisfied are you with your performance?
- Effort: How much effort (both mental and physical) did you have to exert to achieve your level of performance?
- Frustration Level: During the task, did you feel insecure, discouraged, irritated, stressed, and annoyed or secure, gratified, content, relaxed, and complacent?

User Engagement Short Form (1-7 likert scale)

- I lost myself in this experience.
- The time I spent using this application just slipped away.
- I was absorbed in this experience.
- I felt frustrated while using this Application.
- I found this Application confusing to use.
- Using this Application was taxing.
- This Application was attractive.
- This Application was aesthetically appealing.
- This Application appealed to my senses.
- Using Application was worthwhile.
- My experience was rewarding.
- I felt interested in this experience.

USE Questionnaire Short Form (1-7 likert scale)

- It helps me be more effective.
- It is useful.
- It saves me time when I use it.
- It does everything I would expect it to do.
- It is easy to use.
- It is user-friendly.
- It requires the fewest steps possible to accomplish what I want to do with it.
- It is flexible.
- I can use it successfully every time.
- I learned to use it quickly.
- I easily remember how to use it.
- It is easy to learn to use it.
- I quickly became skilful with it.
- I am satisfied with it.
- I would recommend it to a friend.
- It is fun to use.
- It works the way I want it to work.